Sardar Vallabhbhai National Institute of Technology (SVNIT) Surat
Department of Artificial Intelligence
5th Sem , B.Tech. (AI) (2026-27)

Information Security and Cryptography - AI 331
Lab Assignment 3

Scenario *: MedTech Systems* requires a lightweight encryption prototype for a local medical device. The device collects patient identification numbers (e.g., `PATIENT-9842`) and short diagnostic logs (e.g., `PATIENT HAS MILD FEVER`), then encrypts them before passing them through an internal logging network.

Task :

1. Write a function, `brute_force_caesar(ciphertext)` that prints all 25 possible decrypted variations of the output string produced by the Caesar cipher.
2. Implement **Monoalphabetic Substitution Cipher** to encrypt and decrypt sensitive patient data (Patient IDs and diagnostic notes) for a basic medical portal prototype.